Background AI Talent Guard: Predicting & Preventing Employee Attrition and Objectives

GCI FINAL BUSINESS PROPOSAL

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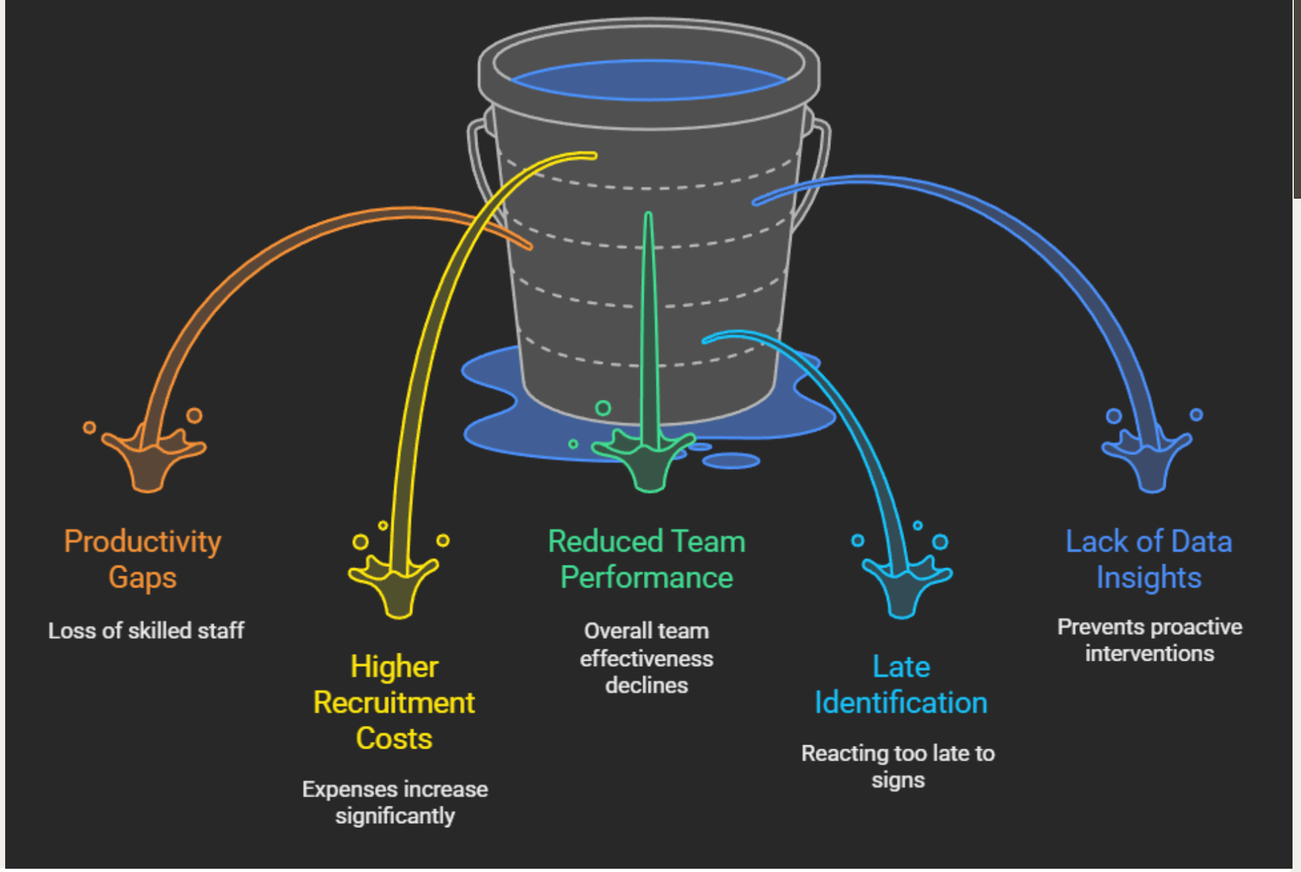
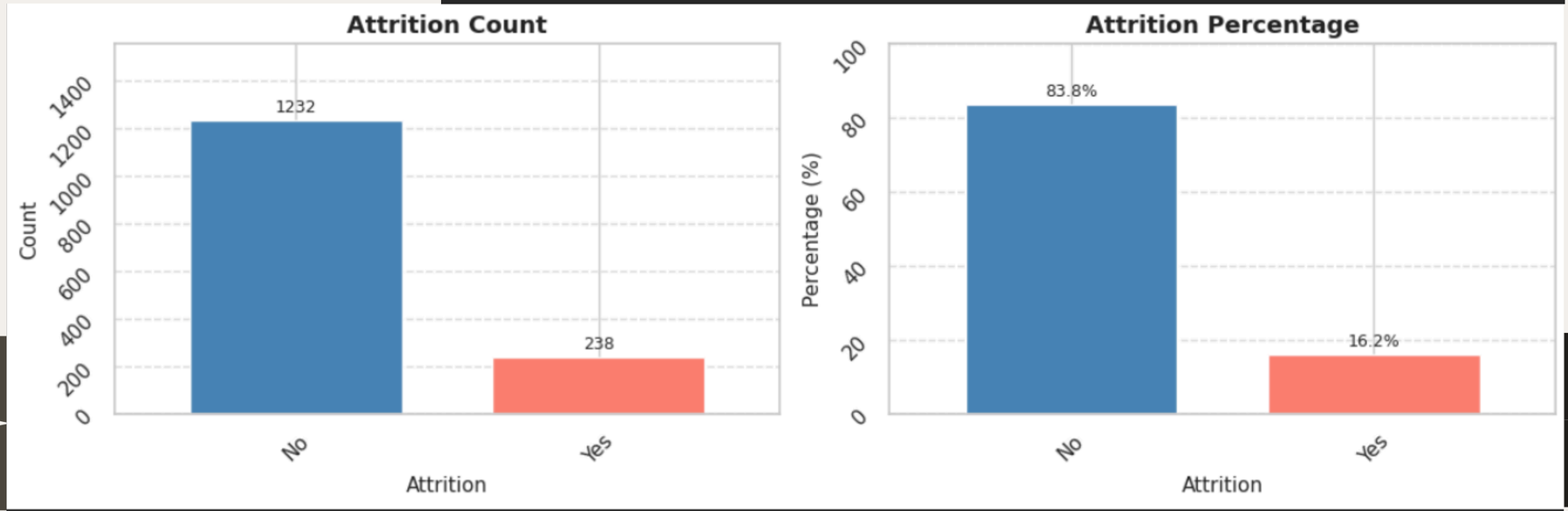
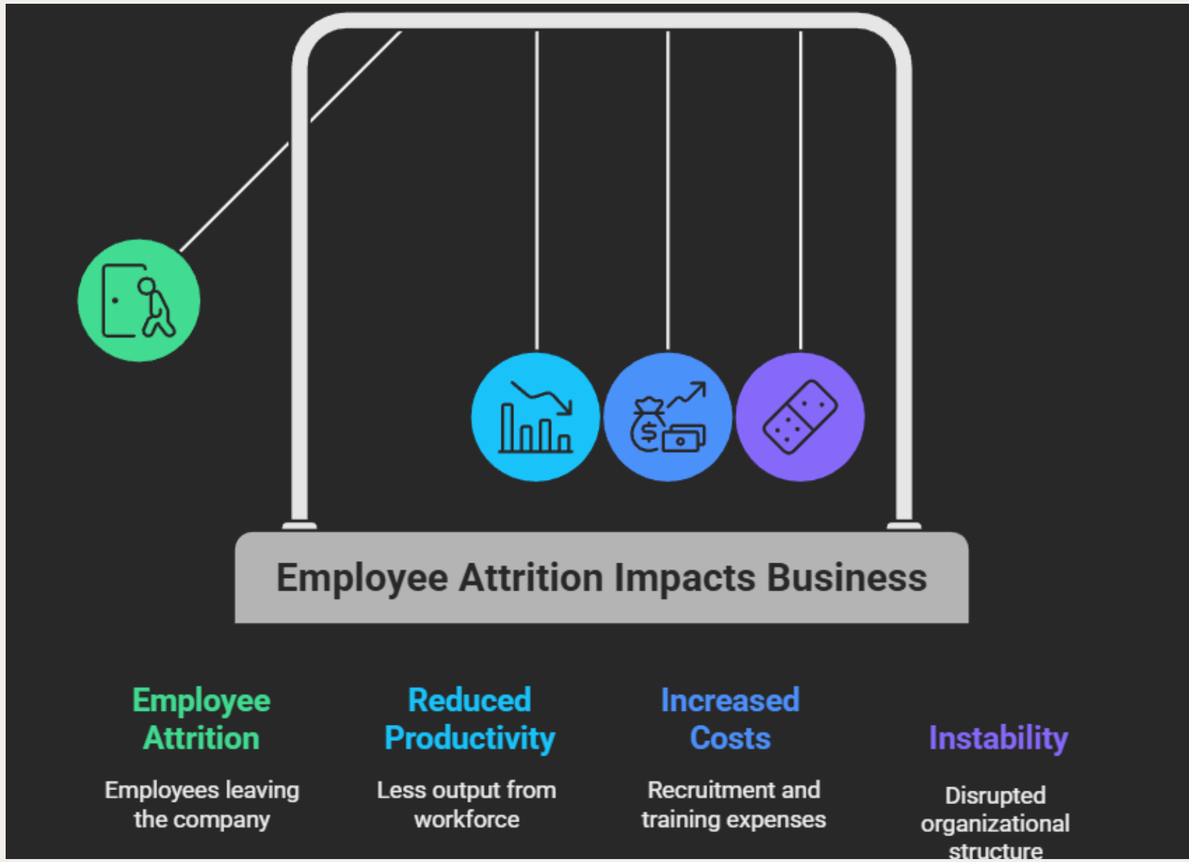
THE PROBLEM


Productivity Gaps
Loss of skilled staff
Higher Recruitment Costs
Expenses increase significantly
Reduced Team Performance
Overall team effectiveness declines
Late Identification
Reacting too late to signs
Lack of Data Insights
Prevents proactive interventions


Attrition Count
Count
0
200
400
600
800
1000
1200
1400
1232
238
No
Yes
Attrition
Attrition Percentage
Percentage (%)
0
20
40
60
80
100
83.8%
16.2%
No
Yes
Attrition

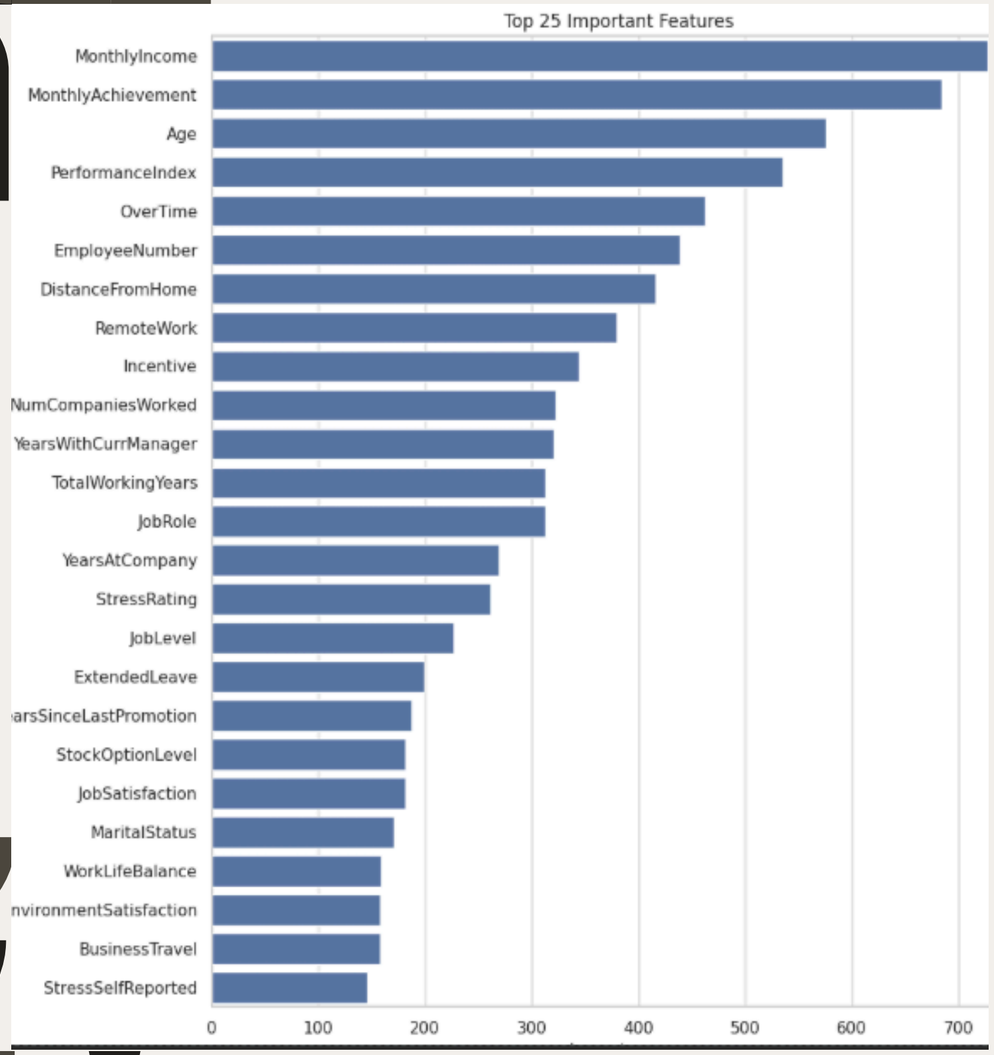
MARKET OVERVIEW

Employee attrition directly affects productivity, recruitment costs, and organizational stability. By predicting which employees are likely to leave, companies can proactively address risks, improve retention, and strengthen workforce planning. This solution enables data-driven decisions, reduces financial losses, boosts employee satisfaction, and ensures long-term business continuity—making it a critical investment for modern HR strategy.



- Losing one employee can cost 30%–200% of their annual salary.
- Turnover impacts productivity, morale, and employer reputation.
- Predictive analytics is no longer optional — it's a competitive advantage.

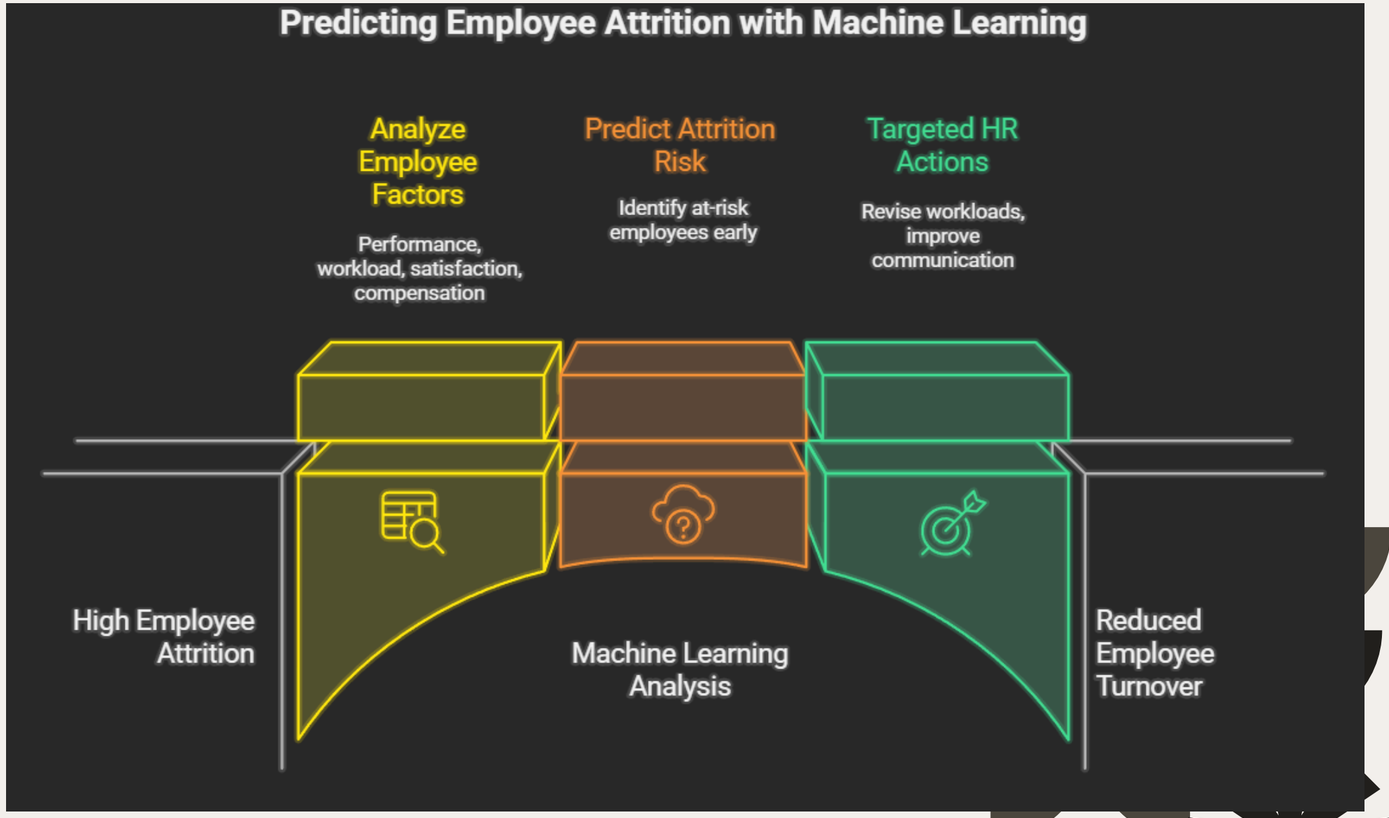
SOLUTION



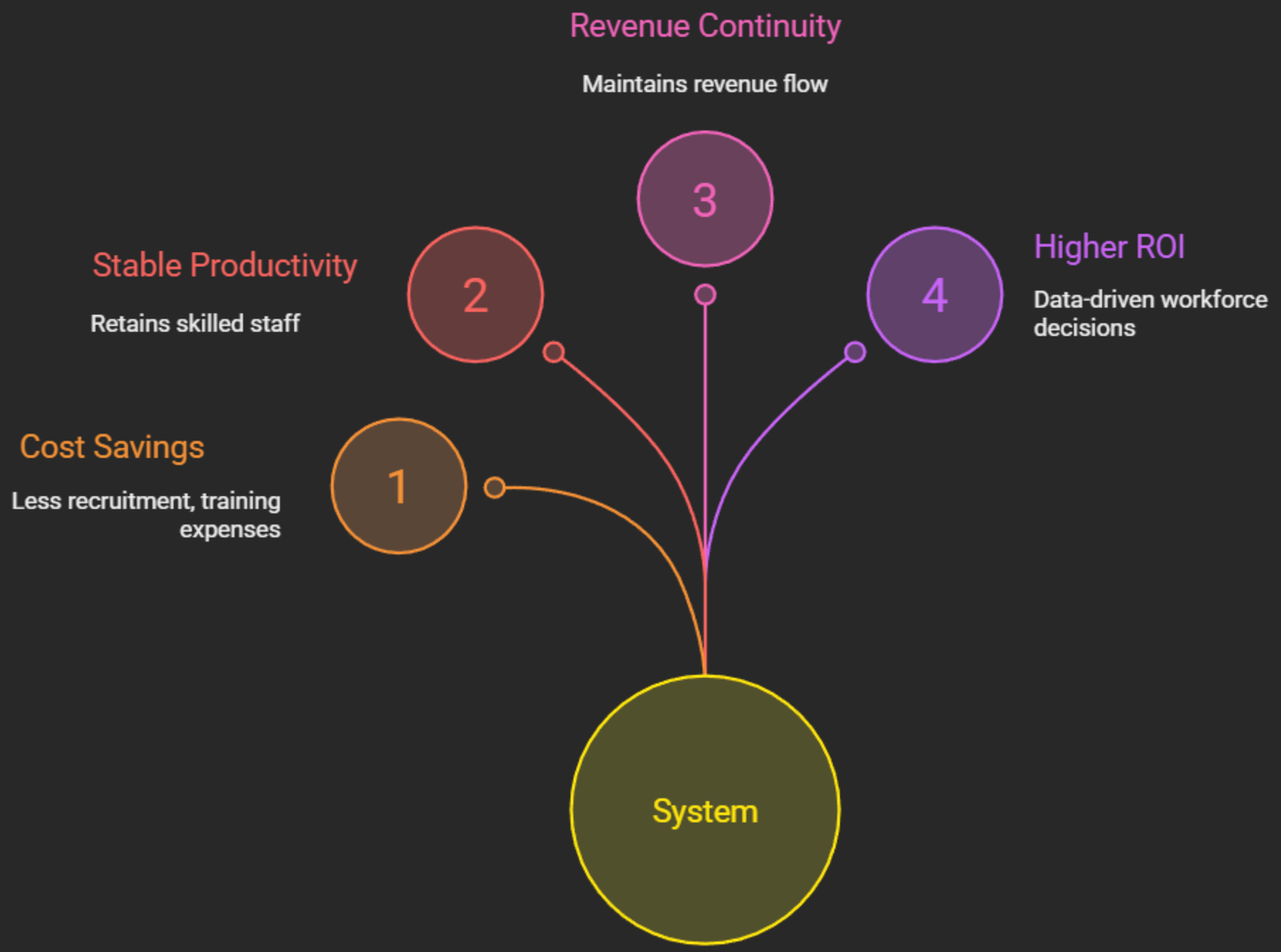
AI TalentGuard

A machine-learning powered attrition prediction system that identifies:

- Who is likely to leave
- Why they might leave
- How to retain them

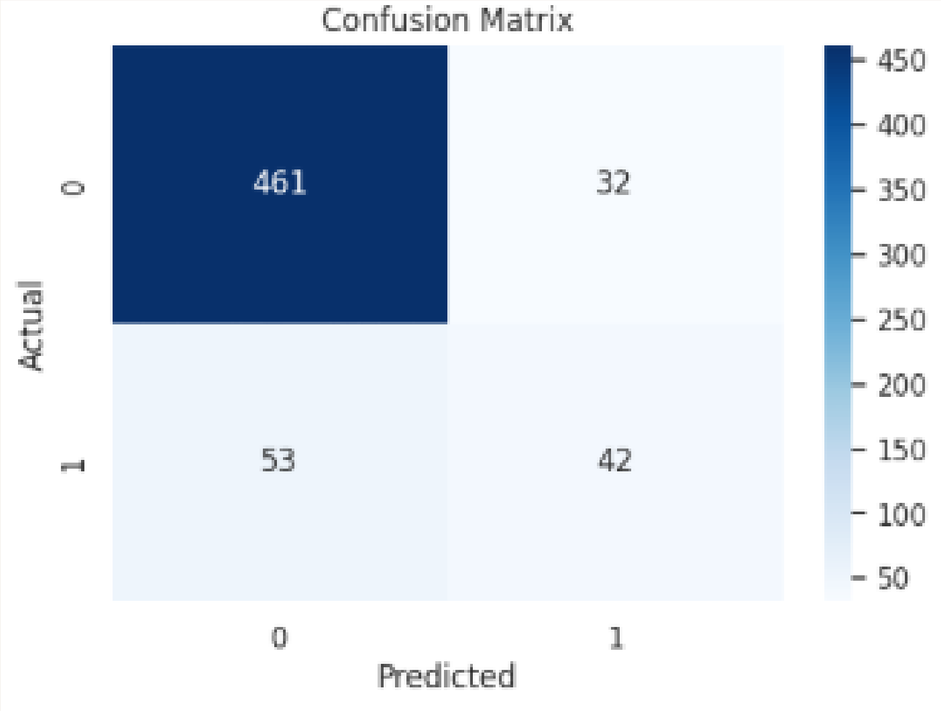


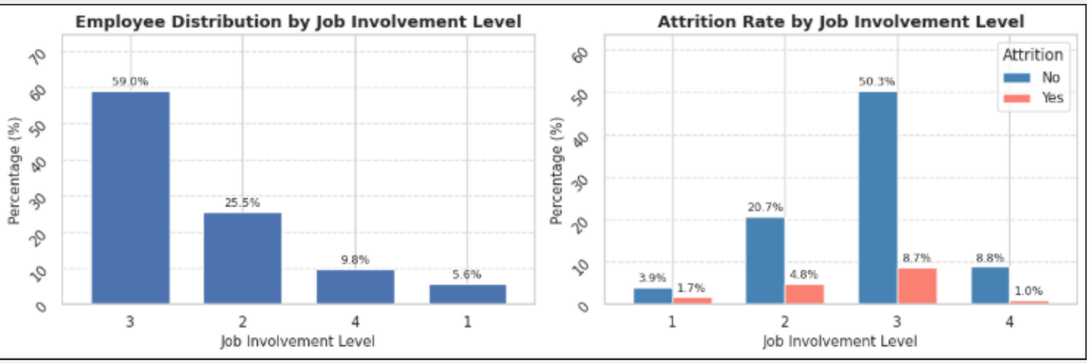
System Benefits

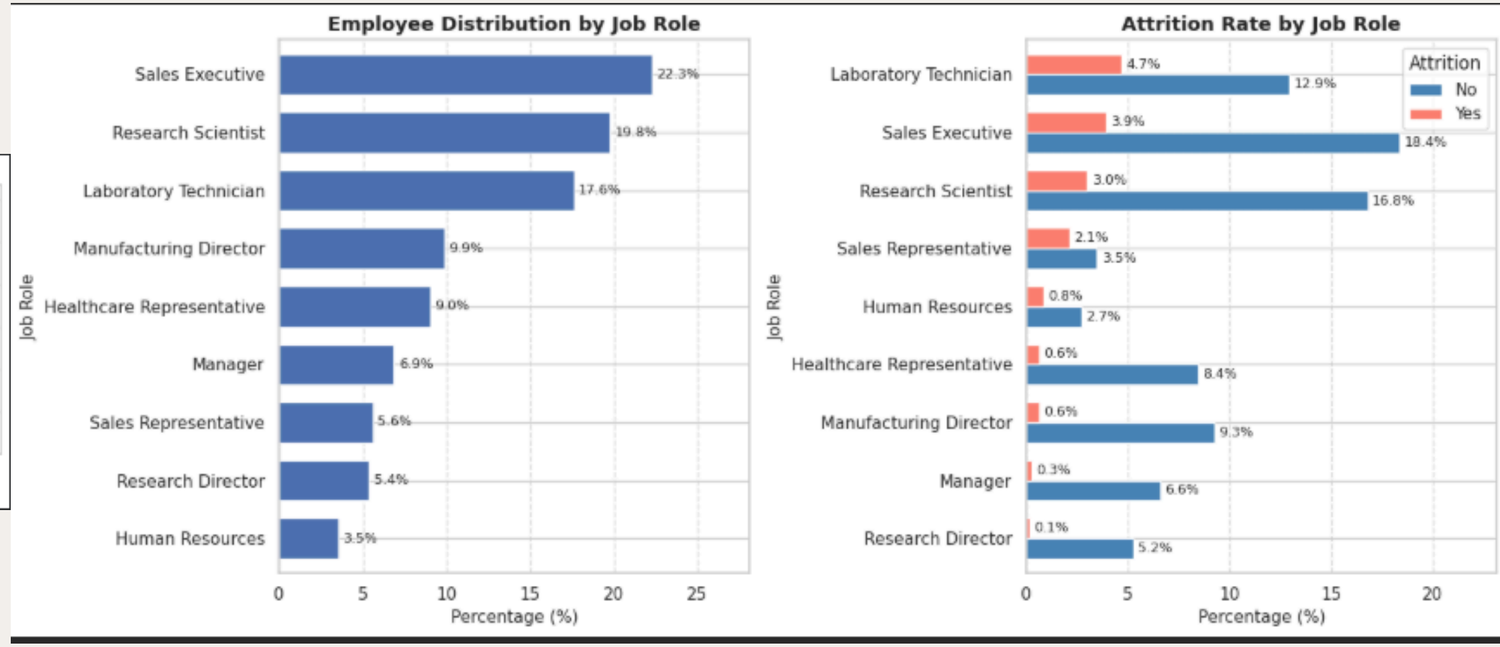


HOW IT WORKS

- CLEAN AND ANALYZE HR DATA
- TRAIN ADVANCED ML MODELS
- PREDICT ATTRITION RISK WITH HIGH ACCURACY
- EXPLAIN RESULTS USING SHAP
- PROVIDE ACTIONABLE RETENTION STRATEGIES







BENEFITS TO YOUR ORGANIZATION

Implementing this system provides organizations with significant benefits by reducing employee turnover and preventing avoidable resignation-related costs. It improves workforce stability, strengthens productivity, and minimizes disruptions caused by talent loss. Through predictive analytics, companies gain clear insights into attrition risks, enabling proactive interventions and smarter HR decision making. This leads to lower recruitment expenses, more efficient resource allocation, and stronger long-term performance, ultimately boosting profitability, operational efficiency, and overall return on investment.

FINANCIAL PROJECTIONS

Employee Loss Impacts Company Finances

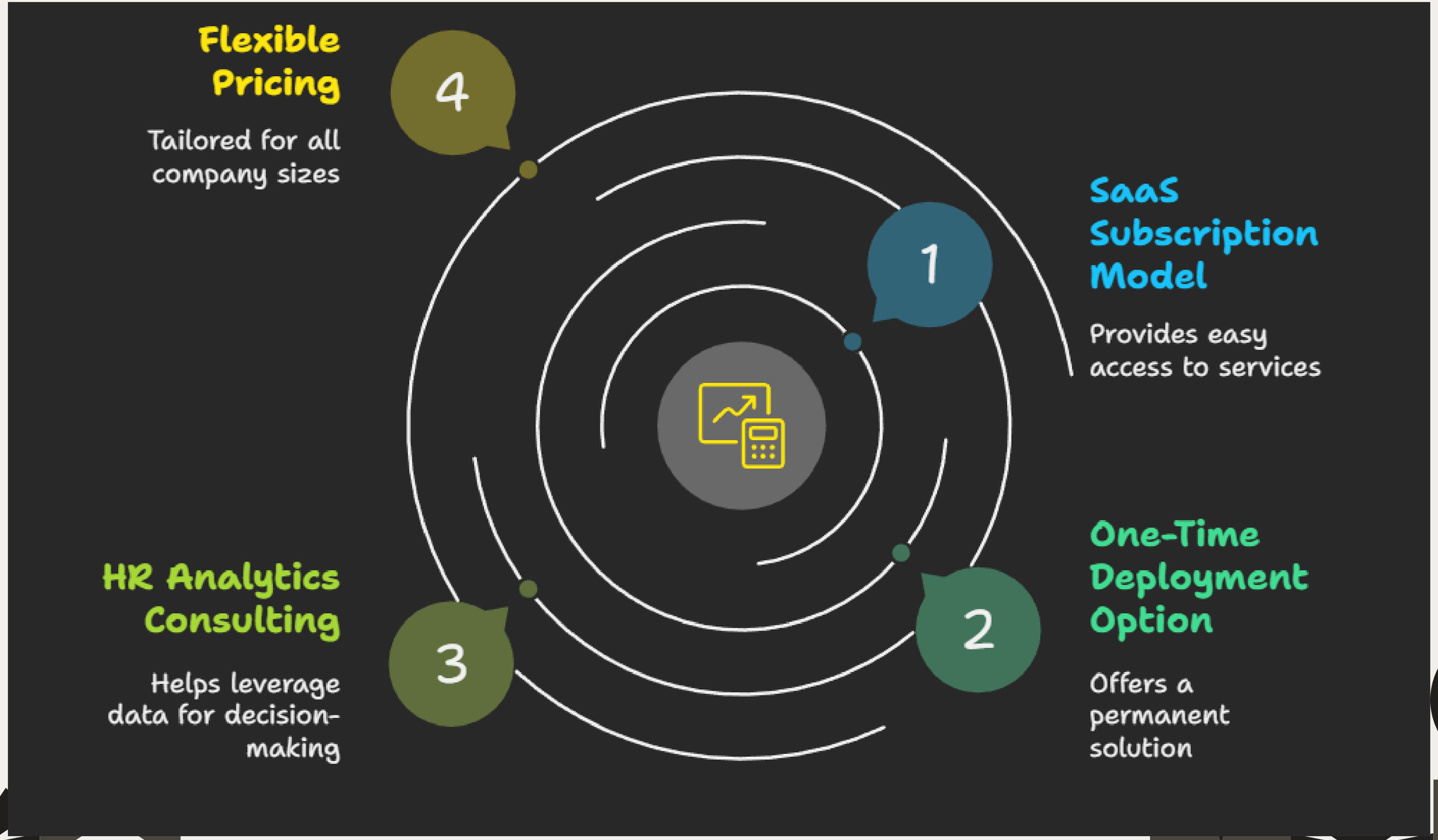
Replacement Costs

$5,000–$20,000 per employee

Annual Loss

$250,000 to $1M+ yearly

BUSINESS MODEL



Call To Action

References

- Romeo, L. N. (2023, March). Cracking the Employee Attrition Problem with Machine Learning.
- Guerranti, F., & Dimitri, G. M. (2023, April). A Comparison of Machine Learning Approaches for Predicting Employee Attrition.